

Part 2: Battery Thermal Model

Revision Notes — Formulas, Reasoning, and FAQ

EV Powertrain Simulator — Thermal Analysis Extension

Prepared as a self-study revision document covering every formula, design decision, and doubt raised while building the model.

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1. Objective and Scope

Part 1 of the EV Powertrain Simulator modeled the **electrical and mechanical** side of an EV: state of charge (SOC) via Coulomb counting, a driving profile with regenerative braking, motor electrical dynamics, and sensor noise/filtering.

Part 2 extends this into the **thermal domain**: how does the battery's temperature evolve during operation, and can different cooling strategies keep it within a safe range? Thermal management is one of the most safety-critical aspects of real EV design — Li-ion cells degrade rapidly, and can become dangerous, above certain temperatures.

The goal was not just to build a working Simulink model, but to derive it from first principles, predict its behavior by hand, and then verify the simulation matches that prediction — the same validation loop used in real engineering work.

2. Physical Concept (Plain-Language Explanation)

Forget equations for a moment. Physically, two things happen inside a battery cell at the same time:

- **Heat is generated** — a battery is not a perfect conductor; it has internal resistance. Whenever current flows through any resistance, energy is converted to heat. This is the exact same reason a phone charger or a resistor gets warm. This is called **Joule heating**.
- **Heat is lost** — the battery sits in air (or a cooling system), and heat naturally flows from hot things to cold things. As the battery heats up, it loses some of that heat to its surroundings via **convection**.

The *difference* between heat generated and heat lost determines whether temperature rises, falls, or stays steady. This is simply **energy conservation** — the most fundamental principle in thermal physics: energy cannot appear or disappear, it only moves around.

Analogy: a cup of coffee at 90°C in a 25°C room loses heat fast; the same coffee at 30°C loses heat slowly. The bigger the temperature difference from the surroundings, the faster heat escapes. That single idea (Newton's Law of Cooling) is the basis of the heat-loss term used below.

3. Governing Equation — Full Derivation

Step 1 — Heat generated (Q_{gen}):

When current I flows through internal resistance R , the power dissipated as heat follows Joule's law (the same $P = I^2R$ used in basic circuits):

$$Q_{gen} = I^2 \times R_{internal}$$

Step 2 — Heat lost to surroundings (Q_{loss}):

Governed by Newton's Law of Cooling: the rate of heat loss is proportional to how much hotter the object is than its surroundings.

$$Q_{loss} = h \times A \times (T - T_{ambient})$$

where $(T - T_{\text{ambient}})$ is the temperature difference driving the heat flow, A is the cell's surface area (more surface = faster heat escape), and h is the convective heat transfer coefficient — a number describing how effective the cooling method is (still air = low h ; a fan = higher h ; liquid cooling = much higher h).

Step 3 — Net heat and rate of temperature change:

Net heat = $Q_{\text{gen}} - Q_{\text{loss}}$. To convert this into a *rate of temperature change*, divide by the cell's thermal mass (mass $\times$ specific heat capacity), which describes how much energy it takes to raise the cell's temperature by 1 K — a bigger or more heat-resistant cell needs more energy to heat up by the same amount.

$$dT/dt = (Q_{\text{gen}} - Q_{\text{loss}}) / (m \times c)$$

Substituting the full expressions gives the complete governing equation:

$$dT/dt = (I^2 \cdot R_{\text{internal}} - h \cdot A \cdot (T - T_{\text{ambient}})) / (m \cdot c)$$

Symbol	Meaning	Unit
I	Current through the cell	A
R_{internal}	Internal resistance	Ω
h	Convective heat transfer coefficient	$W/(m^2 \cdot K)$
A	Cell surface area	m^2
T	Cell temperature	$^{\circ}C$
T_{ambient}	Ambient (surrounding) temperature	$^{\circ}C$
m	Cell mass	kg
c	Specific heat capacity	$J/(kg \cdot K)$

4. Feedback Loop — Why This Model Differs From Part 1

In Part 1's SOC model, current I went in, was integrated, and SOC came out — a one-directional (**open-loop**) calculation. SOC itself never affected how current was calculated.

Here, it is different. To calculate Q_{loss} , the *current temperature* T is needed — but T is exactly the quantity being calculated. The logic forms a loop:

- 1 Current temperature T is used to calculate Q_{loss} .
- 2 $Q_{\text{gen}} - Q_{\text{loss}}$ gives the instantaneous rate of change, dT/dt .
- 3 dT/dt is integrated over time to produce a *new* T .

4 That new T feeds back into step 1, to compute the *next* Q_loss.

This circular dependency — where the output (T) affects a future input to itself — is exactly what **feedback** means in engineering. It is not added complexity for its own sake; it is physically true that a hotter battery loses heat faster, so the system naturally self-corrects toward equilibrium. This is why real batteries heat up fast at first, then level off — an **exponential** response — instead of heating up in a straight line forever.

5. Thermal Time Constant

The rate of settling for this type of first-order feedback system is characterized by a single number, the thermal time constant τ :

$$\tau = (m \times c) / (h \times A)$$

After 1τ , the system has covered roughly 63% of its total temperature change. After 5τ , it is considered fully settled (>99% of the way to steady state). This is why simulation Stop Times were chosen as $5 \times \tau$ or more — running for less time would show an incomplete, misleadingly-still-rising curve instead of the true settled behavior.

6. Model Parameters (Reference 18650 Li-ion Cell)

Part 1's SOC model used a placeholder Coulomb-counting gain (-50) that was never tied to a real battery capacity or internal resistance — it was chosen only to make the SOC curve move at a convenient rate during testing. Back-calculating from it gave an unrealistic $Q \approx 0.00056$ Ah, confirming it was a placeholder, not a real spec.

Rather than inherit an arbitrary number, Part 2 deliberately uses standard reference parameters for a real, well-documented component: a single **18650 Li-ion cell**. This is the same approach a real engineer would take when picking up a prototype with placeholder values — pick a real, citable reference rather than propagate an arbitrary one.

Parameter	Symbol	Value
Internal resistance	R	$0.05\ \Omega$
Cell mass	m	$0.045\ \text{kg}$
Specific heat capacity	c	$900\ \text{J}/(\text{kg}\cdot\text{K})$
Surface area	A	$0.0042\ \text{m}^2$
Ambient temperature	T_{ambient}	25°C
Test current (steady-state tests)	I	$10\ \text{A}$

Three cooling scenarios were evaluated by varying h :

Cooling Method	$h\ (\text{W}/\text{m}^2\cdot\text{K})$	$h\cdot A\ (\text{W}/\text{K})$
None (natural convection)	8	0.0336
Forced air (fan)	25	0.105
Liquid cooling	200	0.84

7. Simulink Implementation — Block by Block

Two versions were built. **Phase 4a** uses a constant current (Step block) to validate against hand calculations. **Phase 4b** uses a realistic drive-cycle current (via a From Workspace block) to test dynamic behavior. Both share the same core structure:

Block(s)	Function
Step / From Workspace	Supplies current I — either constant (4a) or time-varying (4b)

Math Function (u^2)	Computes I^2
Gain = R_{internal} (0.05)	Multiplies I^2 by R to get Q_{gen}
Constant = 25 (T_{ambient}) + Sum block	Computes $(T - T_{\text{ambient}})$, with T fed back from the Integrator
Gain = $h \cdot A$	Multiplies $(T - T_{\text{ambient}})$ by $h \cdot A$ to get Q_{loss}
Sum block (NET HEAT)	Computes $Q_{\text{gen}} - Q_{\text{loss}}$
Gain = $1/(m \cdot c)$	Divides net heat by thermal mass to get dT/dt (Gain blocks only multiply, so the reciprocal $1/(m \cdot c)$ is entered directly)
Integrator (1/s)	Integrates dT/dt to get T . Initial condition set to 25°C (ambient)
Feedback wire	Routes the Integrator's output T back to the $(T - T_{\text{ambient}})$ Sum block, closing the loop
Scope	Displays T over simulated time

Drive cycle definition (Phase 4b)

Defined in MATLAB and imported into Simulink via a From Workspace block reading a variable `drive_cycle` (a two-column time/current matrix):

```
time = [0; 20; 20; 60; 60; 150; 150; 170; 170; 200];
current = [0; 0; 15; 15; 5; 5; -8; -8; 0; 0];
drive_cycle = [time, current];
```

Phase	Time (s)	Current (A)	Represents
Idle	0–20	0	Vehicle stopped
Accelerate	20–60	15	Hard acceleration
Cruise	60–150	5	Steady speed
Regenerative brake	150–170	–8	Braking, energy recovery
Idle	170–200	0	Vehicle stopped

8. Worked Hand Calculation (Numeric Walkthrough)

Before building anything in Simulink, the model's behavior was predicted by hand using simple round numbers, to see the shape of the feedback response clearly. Assumptions: $Q_{\text{gen}} = 5 \text{ W}$ (constant), $h \cdot A = 0.5 \text{ W/K}$, $m \cdot c = 40 \text{ J/K}$, $T_{\text{ambient}} = 25^\circ\text{C}$, starting $T = 25^\circ\text{C}$, time step = 10 s.

Time (s)	T (°C)	Q_loss (W)	Net heat (W)	dT/dt (K/s)	New T (°C)
0	25.00	0.000	5.000	0.1250	26.25
10	26.25	0.625	4.375	0.1094	27.34
20	27.34	1.170	3.830	0.0958	28.29

Notice each step's temperature rise is smaller than the previous one ($1.25 \rightarrow 1.09 \rightarrow 0.96^\circ\text{C}$) — this is the feedback loop visibly slowing the rise as T climbs and Q_{loss} grows.

Steady state occurs when $Q_{\text{gen}} = Q_{\text{loss}}$ exactly:

$$5 = 0.5 \times (T_{\text{final}} - 25) \Rightarrow T_{\text{final}} = 35^\circ\text{C}$$

Time constant for this test case: $\tau = 40/0.5 = 80 \text{ s}$, so full settling (5τ) occurs by roughly 400 s. This hand-calculated curve (25°C rising smoothly and flattening at 35°C) was later matched almost exactly by the actual Simulink simulation, confirming the model was built and wired correctly.

9. Results — Steady Current Tests (10 A constant)

Cooling Method	Predicted T_{final}	Simulated Result	Verdict
None (natural convection)	173.8°C	Still climbing steeply, never settles safely	Dangerous
Fan (forced air)	72.6°C	$\approx 70^\circ\text{C}$ and still rising at $t=1100\text{s}$	Exceeds safe limit
Liquid cooling	31.0°C	$\approx 31^\circ\text{C}$, fully settled and flat	Safe

The liquid-cooling case matched the hand-calculated prediction (31.0°C) almost exactly, validating the correctness of the Simulink build.

10. Results — Realistic Drive-Cycle Tests

Cooling Method	Peak Temp	Temp at $t=200\text{s}$	Behavior
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None	≈39.3°C	≈38.8°C	Continuously rising; no recovery even during cruise/idle
Fan	≈36.8°C	≈36.0°C	Nearly flat during cruise; mild net rise overall
Liquid	≈32.5°C	≈26.5°C	Temperature tracks each phase; clear recovery during cruise/braking/idle

With liquid cooling, the model also revealed that **regenerative braking still generates measurable heat** — heat generation depends on I^2 (current magnitude), not direction, so an 8 A braking current produces more heat than the 5 A cruise current, despite being a charging rather than a discharging event. This is a genuine, non-obvious insight the model surfaced on its own.

11. Key Engineering Finding

Across both steady-current and realistic drive-cycle testing, **a single 18650 cell relying on natural convection or simple fan cooling cannot safely dissipate the heat generated under sustained or repeated high-current draw**. Only liquid cooling kept the cell within a safe temperature band (roughly 25–33°C) in every test condition.

This result was not assumed going in — it fell out of the model itself, derived entirely from first principles. It matches why virtually all modern EV battery packs use liquid cooling systems rather than passive or air-based approaches.

12. Frequently Asked / Doubts Clarified

Q: Is this a real-time model, running like it would on actual EV hardware?

A: No. This is an offline simulation model, run in MATLAB/Simulink on a laptop. Simulated time can run faster or slower than real clock time (a 500-second scenario can compute in a couple of seconds). A real-time model would instead run on embedded hardware (like a Battery Management System microcontroller), compute at the exact pace of real time, and use live sensor inputs. Building and validating offline first, as done here, is the standard industry progression before anything is deployed as real-time control logic.

Q: Why does the model use 'placeholder' values instead of Part 1's real numbers?

A: Part 1's SOC model never defined a physical internal resistance or realistic capacity — its gain value was chosen only to make the SOC curve move at a testable rate, not to represent a real cell. Rather than inherit that placeholder, Part 2 deliberately switched to standard, real, documented 18650 cell parameters, making the thermal analysis defensible and citable.

Q: Why is this a 'feedback loop' and not just a normal calculation chain?

A: Because Q_{loss} depends on the current temperature T , and T is exactly the quantity being solved for. The Integrator's output is wired back into the Q_{loss} calculation, creating a closed loop where the system's own output influences its future input — this is what produces the exponential rise-and-settle curve shape, rather than unbounded linear growth.

Q: Why did the '5 x time constant' rule matter for choosing Stop Time?

A: A system only reaches roughly 99% of its final value after 5 time constants (5τ). Running a simulation for less time shows an incomplete curve that looks like it is still climbing when it is actually approaching a stable value — this caused early confusion when a 10 s or 500 s Stop Time cut the curve off before it had a chance to visibly flatten.

Q: Why did regenerative braking still increase temperature instead of cooling the battery?

A: Heat generation is proportional to I^2 , which only depends on current magnitude, not its sign (charging vs. discharging). An 8 A braking current generates more heat than a 5 A cruising current, even though braking is 'putting energy back' into the battery electrically.

Q: Why are there two separate Simulink files (Phase 4a and 4b) instead of one?

A: Phase 4a (constant current, Step block) exists specifically to validate the model against hand calculations under simple, predictable conditions. Phase 4b (drive-cycle, From Workspace block) tests the same validated model under realistic, dynamic conditions. Keeping them separate preserves both the validation evidence and the realistic-behavior demonstration.

Q: What does 'Part 2, Phase 1' vs 'Phase 4' mean — which naming is correct?

A: Both refer to the same work, at different levels of the project's hierarchy. Internally, this is 'Part 2' of the overall project (Part 1 = electrical/mechanical: SOC, driving profile, motor). Within Part 2, the steady-current model is its 'Phase 1' and the drive-cycle model is its 'Phase 2'. In the single flat numbering used across the whole repository's file names and images, this same work is labeled 'Phase 4' (continuing after Part 1's Phases 1–3). The physics and files are identical either way — only the labeling scheme differs.

Q: Why does a Gain block need $1/(m \cdot c)$ instead of just $m \cdot c$?

A: Simulink's Gain block only multiplies its input; it cannot divide. To perform the division 'net heat $\div (m \cdot c)$ ', the reciprocal $1/(m \cdot c)$ is calculated first and entered directly as the Gain value, so multiplying by this reciprocal has the same effect as dividing by $(m \cdot c)$.

13. Limitations and Future Work

- **Single-cell scope** — this model represents one reference cell, not a full pack. A pack-level model would need cell-to-cell heat transfer, uneven current sharing across parallel strings, and pack-level cooling plate geometry.
- **Irreversible heat only** — the reversible (entropic) heat term, which changes sign between charge and discharge, was omitted. It is typically smaller than I^2R losses under normal driving but would add fidelity for high-precision analysis.
- **Simplified drive cycle** — a single idealized cycle (accelerate–cruise–brake–idle) was hand-defined. Feeding this thermal model directly from Part 1's motor/current output would test it against the project's own simulated driving behavior instead.
- **Constant cooling coefficient** — real cooling systems (especially fans) often have coefficients that vary with airflow rate, itself dependent on speed or a control system; held constant here for simplicity.

14. Quick-Reference Formula Sheet

Quantity	Formula
Heat generated	$Q_{\text{gen}} = I^2 \times R_{\text{internal}}$
Heat lost	$Q_{\text{loss}} = h \times A \times (T - T_{\text{ambient}})$
Rate of temperature change	$dT/dt = (Q_{\text{gen}} - Q_{\text{loss}}) / (m \times c)$
Full governing equation	$dT/dt = (I^2R - h \cdot A \cdot (T - T_{\text{ambient}})) / (m \cdot c)$
Thermal time constant	$\tau = (m \times c) / (h \times A)$
Steady-state temperature	$T_{\text{final}} = T_{\text{ambient}} + Q_{\text{gen}} / (h \times A)$
Full settling time (~99%)	$t \approx 5\tau$

End of revision notes. Refer to the project README and Simulink files (battery_thermal_model.slx, battery_thermal_model_phase2.slx) for the working implementation.